**EXAM SHIELD**

# Computer Science Question Bank

Algorithms, data structures, number systems and systems.

13 questions: 8 multiple choice, 3 fill in the gap and 2 written. Total 19 marks.
Edition: with answers (staff copy, ready to import into Exam Shield).

## How this paper is laid out

Questions are numbered in one sequence. Multiple-choice options are lettered with a full stop; each multiple-choice question is worth one mark unless stated.

Fill-in-the-gap questions show each blank as a line; they earn one mark per blank, shown in square brackets, and are marked automatically.

Written questions show their marks in square brackets and are marked by an examiner.

To use this paper in Exam Shield, open an exam, choose Questions and then Import from PDF, and upload this file. Review the preview and import.

# Section 1: Computer Science

Algorithms, data structures, number systems and systems.

**1. What is the time complexity of binary search on a sorted array?**

A. O(1)
B. O(log n)
C. O(n)
D. O(n log n)

Answer: B

**2. Which data structure follows Last-In, First-Out (LIFO) order?**

A. Queue
B. Linked list
C. Binary search tree
D. Stack

Answer: D

**3. What is the average-case time complexity of quicksort?**

A. O(n log n)
B. O(n^2)
C. O(n)
D. O(log n)

Answer: A

**4. What is the binary representation of decimal 10?**

A. 1100
B. 0110
C. 1001
D. 1010

Answer: D

**5. Which algorithm finds shortest paths in a graph with non-negative edge weights?**

A. Depth-first search
B. Dijkstra's algorithm
C. Kruskal's algorithm
D. Prim's algorithm

Answer: B

**6. What does a compiler do?**

A. Translates source code into machine or target code
B. Executes source code one line at a time
C. Compresses files for storage
D. Translates machine code back into source code

Answer: A

**7. What is the average time to look up a key in a hash table?**

A. O(1)
B. O(n^2)
C. O(n)
D. O(log n)

Answer: A

**8. Which of these is NOT a core operating system function?**

A. Compiling user programs
B. Process scheduling
C. Memory management
D. File management

Answer: A

**9. Breadth-first search uses a _____ to hold the nodes to visit, while depth-first search uses a _____ or recursion. [2 marks]**

Answer: queue | stack

**10. A tree in which every node has at most two children is called a _____ tree. [1 mark]**

Answer: binary

**11. A problem is NP-complete if it is in NP and every problem in NP can be _____ to it in polynomial time. [1 mark]**

Answer: reduced

**12. Explain recursion and the role of the base case. Give a simple example. [3 marks]**

**13. Compare a stack and a queue, giving one real-world use for each. [4 marks]**